Semantic-Aware Scheduling for Minimizing Age of Informative Data in WBAN-Based Health Monitoring Systems

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***Abstract*—Since outdated information can lead to critical situations in health monitoring systems, recent schedulers have adopted the Age of Information (AoI) as a key decision metric. However, traditional AoI-based schedulers penalize all data uniformly over time, even when successive packets contain redundant content, failing to prioritize informative data. Although semantic-aware AoI schedulers address this limitation, they often become overly sensitive to minor variations in content, neglecting to quantify data criticality and ultimately behaving like traditional AoI-based schedulers. To overcome this issue, in this article, a new semantic-aware AoI scheduler is proposed for wireless body area network (WBAN)-based health monitoring systems. The scheduling objective focuses on minimizing the weighted average AoI, where the weights are based on deviations from historical data, ensuring timely emergency detection. Additionally, an effective optimization algorithm is developed and applied to achieve this goal. Simulation results show that the proposed scheduler improves system performance across key metrics, including average AoI, weighted average AoI, and throughput, ensuring the freshness of informative data in WBAN-based health monitoring systems.**

***Index Terms*—Age of informative data, deep reinforcement learning (DRL), self-adaptive greedy scheduling, semantic-aware scheduling.**

I. Introduction

WIRELESS body area network (WBAN) is a specialized network that supports on-body communication between a coordinator and wearable sensors. WBANs are mainly used for building health monitoring systems wherein medical sensors collect various physiological signals from the body and forward them to the coordinator. Technically, the coordinator is responsible for aggregating the received physiological data as well as allocating radio resources (i.e., timeslots) to each sensor. According to IEEE 802.15.6 [1], the WBAN communication standard, each sensor shares a transmission channel in the 2.4-GHz industrial, scientific, and medical (ISM) band. Hence, radio resources are not sufficient and thus the coordinator should carefully distribute timeslots taking into account the technical requirements of WBANs.

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A key consideration for resource allocation in WBANs is channel fading due to frequent body movements. The human body itself can be an obstacle that mitigates the strength of the wireless signal between the coordinator and body sensors. If fading leads to packet errors and losses, retransmissions become necessary to ensure data integrity. These retransmissions consume additional bandwidth and time resources, significantly reducing the system throughput. For this reason, conventional scheduling techniques have focused on improving the system throughput (i.e., maximizing the amount of collected information per unit time) by recognizing dynamic channel conditions and allocating timeslots accordingly [2]. For example, in [3], [4], [5], and [6], the coordinator (i.e., scheduler) allocates timeslots to each source at the peak point with the best link quality according to the gait cycle. If the link quality between the coordinator and the source is poor, the resource allocation is postponed.

However, some researchers claim that conventional performance metrics (e.g., throughput and packet loss) are not suitable for evaluating efficiency of schedulers in real-time monitoring systems. As seen above, existing schedulers preferentially allocate timeslots to sources with a high probability of successful transmission, or adjust the sending sequence of each source after recognizing movement patterns to improve the system throughput. However, there is a reasonable possibility that the status of sources with relatively low link quality may not be updated for a long period of time (despite the transmission is actually possible). If the status of a source has not been updated on the receiver side for a while, the last sampled data from the source becomes outdated and no longer useful.

Since outdated information can lead to emergencies, especially in health monitoring systems, recent studies have started to introduce a new metric, namely Age of Information (AoI) [7] in designing of schedulers. AoI is an application layer (APP) metric to capture the freshness of information, which is defined as the time elapsed since the latest data was generated and received at the destination. By definition [7], the AoI of a source i increases linearly over time at the coordinator (i.e., receiver) side and resets to 1 when the coordinator allocates a timeslot to source i and successfully receives a new sample from that source. Following this rule, the scheduler prioritizes sources with higher AoI to minimize the average AoI of the whole system [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25].

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However, AoI may not be an ideal metric for scheduler design in health monitoring systems, as it can impose unnecessary penalties even when the information content of consecutive packets is identical or nearly unchanged. For example, if the data observed from a heart rate sensor remains at the same level (i.e., steady state) over N sampling periods, there is no need to quickly update the next state. If there is a significant change in the last received sample, it should be considered an emergency (e.g., heart attack) and the coordinator must obtain further information as quickly as possible. However, traditional AoI-based schedulers, which penalize AoI solely based on time, often prioritize sources with high AoI, regardless of the actual relevance or usefulness of the data. Consequently, they struggle to quickly and consistently detect emergencies, where timely and contextually relevant data are critical.

To ensure timely and continuous emergency detection, the scheduler should make scheduling decisions by considering both the age of the sampled data and its contextual changes. Semantic communication techniques are particularly well-suited for this purpose, as both semantic communication and AoI minimization share the common goal of efficient information transmission. By focusing on transmitting only informative data—defined as newly received data that shows a significant deviation from previously observed data—semantic communication helps reduce the age of such data. Leveraging this synergy, various semantic-aware AoI schedulers have been proposed to optimize the age of informative data across different communication systems [23], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38].

Unlike traditional AoI-based schedulers, semantic-aware AoI schedulers measure the time since the sampled data at the source last matched the data stored at the receiver. Following this rule, AoI increases only when changes occur in the sampled data, ensuring that informative data are prioritized for transmission. However, in health monitoring systems, where physiological data are collected at very short intervals and vital signs fluctuate continuously, successive samples rarely have identical values [39]. As a result, existing semantic-aware AoI schedulers often become overly sensitive to minor variations in the sampled data (i.e., they cannot quantify how much the data has changed) and ultimately behaving similarly to traditional AoI-based schedulers.

To address this limitation, this article introduces a new semantic-aware AoI scheduler tailored for WBAN-based health monitoring systems, focusing on minimizing the age of informative data. To overcome the over-sensitivity issue, the problem is reformulated as minimizing the weighted average AoI, where the weights are determined by the standard deviation from historical data. Additionally, an effective optimization algorithm is proposed to solve this scheduling problem. This approach enhances data freshness and enables timely, continuous detection of emergency situations in health monitoring systems. The main contributions of this article are summarized as follows.

1) This article proposes a new semantic-aware AoI scheduler aimed at minimizing the age of informative data, allowing the receiver to promptly and continuously detect emergency situations in WBAN-based health monitoring systems. To achieve this, a new notion is introduced: newly received data that shows a significant deviation from previously observed data is referred to as informative data.
2) To enable semantic communication in WBAN-based health monitoring systems, this article introduces a new control layer between the medium access control (MAC) and APP layers, facilitating efficient semantic-aware communication. The proposed layer complies with the IEEE 802.15.6 standard, enhancing its adaptability to real-world applications.
3) To address the over-sensitivity issue in existing semantic-aware AoI schedulers, the problem of minimizing the age of informative data is reformulated as minimizing the weighted average AoI, with the weight defined as the standard deviation from historical data. This approach quantifies data criticality, thereby enhancing the detection of emergencies.
4) To minimize the weighted average AoI, the proposed semantic-aware AoI scheduler utilizes a greedy heuristic algorithm. This approach is driven by the need to dynamically adjust the scheduling policy based on contextual fluctuations in the sampled data. Given the difficulty of predicting future states due to varying information content over time, decision-making based solely on the current state is deemed optimal. The effectiveness of the proposed algorithm is validated through comprehensive simulations.

II. Background and Motivation

In this section, traditional AoI-based schedulers proposed in various communication systems are first reviewed. Next, the background that led to the development of semantic-aware AoI schedulers is explored. Finally, the technical challenges faced by existing semantic-aware AoI schedulers are discussed, along with the key motivations for this work.

A. Traditional AoI-Based Schedulers

Most current AoI schedulers concentrate on single-hop wireless networks, where N sources collect data samples and transmit them to a base station (BS) over a shared wireless medium. These systems typically operate within fixed-length timeslots, and the BS allocates slots to individual sources. The main goal of conventional AoI schedulers is to develop an optimal scheduling policy to minimize the average AoI across the network [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. Typically, they achieve this by formulating the AoI minimization problem as an objective function and iterating through different transmission schedules to find the optimal sending sequence that minimizes this function.

Despite these methods, most of them do not enforce specific AoI constraints, making it challenging to meet the stringent AoI requirements of applications that rely on real-time updates, such as health monitoring or autonomous driving. To address this issue, deadline-aware schedulers have been

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introduced [40], [41], [42], [43], [44]. These schedulers ensure that AoI deadlines are met by designing cyclic scheduling strategies that determine a nearly uniform or precise transmission order for each source, aiming to minimize the occurrence of deadline violations. However, finding an optimal solution becomes impractical in real-world communication networks with limited resources. Instead of pursuing the optimal solution, researchers have developed a lower bound for the objective function, which serves as a reference for evaluating AoI schedulers [45], [46], [47], [48]. They argue that a scheduler can be considered optimal as long as the objective function stays within this lower bound.

Several approaches have been proposed to refine the standard AoI metric, introducing alternatives like Age of Synchronization (AoS) [49] and effective AoI (EAoI) [50]. AoS, applied in distributed content caching, behaves differently from AoI by staying at zero when the source lacks data to transmit. However, once the source starts gathering data, AoS grows linearly over time. EAoI, designed for proactive information update scenarios, operates in systems where a server serves multiple users and transmits data only upon request. Unlike AoI, EAoI remains zero until a user sends a request, after which it increases linearly as the server responds to the query. Follow-up studies utilizing AoS and EAoI have continued to emerge in recent years, further exploring and expanding on these concepts [51], [52], [53], [54].

B. Semantic-Aware AoI Schedulers

As researchers have studied AoI more deeply, they realized it has some limits in measuring how fresh information really is. The issue is that AoI only considers the time since data was created, without accounting for whether the information is still useful. For example, if new data are similar and closely related to previous data, then from the destination's perspective, the older data may still be considered fresh, even if the AoI is high for that source. To address this, researchers are focusing on measuring freshness based on the usefulness of the information content. This has led to the development of semantic-aware AoI schedulers, which prioritize scheduling not only based on the age of the sampled data but also on its relevance to previously collected data [55].

In this context, the Age of Correlated Information (AoCI), introduced in [56], describes the freshness of correlated updates. AoCI increases linearly with time and decreases only when new, integrated information is generated. For instance, in wireless sensor networks where sensors collect spatially correlated updates or exchange information locally, AoCI decreases not only when new data are received directly from the source but also when data are received from spatially correlated sources. Thus, AoCI captures both the time elapsed and the value of information exchanged between related sensors. Recently, there has been a surge in the development of semantic-aware AoI schedulers that leverage the AoCI metric [23], [26], [27], [28], [29], [30], [31], [32].

Maatouk et al. [33] introduced a new metric named Age of Incorrect Information (AoII). This metric quantifies the elapsed time since the content of the sample at its source last matched the content of the sample stored at the destination. By definition, AoII increases linearly over time but decreases in two situations: 1) when the destination receives a new sample and 2) when the destination is informed that its stored sample matches the latest sample at the source. In recent years, various semantic-aware AoI schedulers utilizing the AoII metric have been proposed, further exploring and expanding upon these concepts [34], [35], [36], [37], [38].

C. Limitations and Opportunities

Over-Sensitivity Problem: As highlighted in the literature review, various semantic-aware AoI schedulers have been proposed to minimize the age of informative data in wireless communication systems. By utilizing improved semantic-aware age metrics (e.g., AoII), these schedulers can enhance resource efficiency and effectively minimize the age of informative data. Specifically, AoII increases only when newly sampled data at the source differs from the data stored at the destination. However, it is less effective in health monitoring systems, where physiological data are sampled at very short intervals and continuous fluctuations in vital signs mean successive samples rarely have identical values. In other words, AoII is too sensitive to the information content, often imposing penalties for minor variations and ultimately behaving like the original AoI metric, with penalties increasing over time.

Proposed Approach Versus Existing Approaches: To address the over-sensitivity issue, the proposed semantic-aware AoI scheduler shifts the focus to minimizing the weighted average AoI across the system. Unlike traditional methods, this approach leverages weights based on the standard deviation from historical data, enabling a more context-aware evaluation of what qualifies as informative data. By employing a greedy heuristic optimization algorithm to minimize the weighted average AoI, the proposed scheduler effectively mitigates over-sensitivity, enhances the freshness of informative data, and facilitates prompt, continuous detection of emergency situations in health monitoring systems.

III. System Model and Problem Statement

A. Target Model

As shown in Fig. 1, the system comprises a coordinator and multiple source nodes organized in an one-hop star topology. Each sensor is placed on a different body part and collects physiological data based on its specific application type, such as electroencephalography (EEG), electrocardiography (ECG), or elektrokardiogramm (EKG). The coordinator is responsible for aggregating data received from each source node. The system uses a time division multiple access (TDMA) scheme for channel access, where each source node is assigned specific timeslots for data transmission. This setup adheres to the IEEE 802.15.6 standard, which is designed for WBANs, providing reliable communication for low-power devices. In IEEE 802.15.6, the coordinator plays a crucial role in managing the network and ensuring efficient communication. One of its key functions is to provide scheduling information to source nodes. This is accomplished through the periodic transmission

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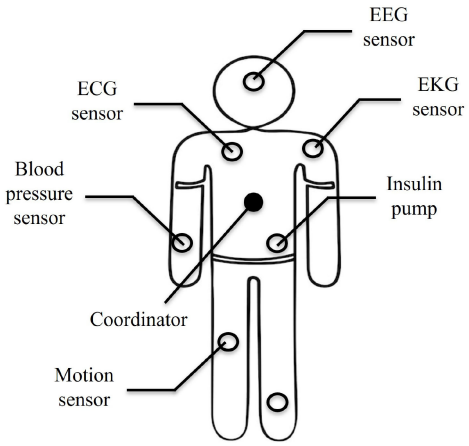


Fig. 1. Envisioned WBAN topology model.

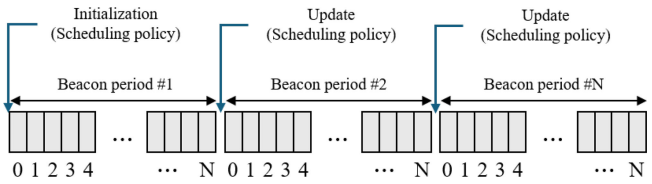


Fig. 2. Illustration of dynamic adjustments in scheduling policy at each beacon interval.

of beacon frames, which contain vital information about the network's timing and resource allocation. The beacon frame contains a TDMA schedule that specifies the exact timeslots assigned to each source node. As illustrated in Fig. 2, the coordinator can update the scheduling information in subsequent beacon frames as needed. This allows for dynamic adjustments to the schedule in response to changing network conditions.

It is assumed that each source collects a new sample at the beginning of each timeslot and transmits the latest data in its buffer at the scheduled timeslots. The coordinator makes scheduling decisions for source i at time $t = 0, 1, 2, \ldots,$ denoted by $\mu_i(t) \in \{0, 1\}$. If source i is scheduled at time t, then $\mu_i(t) = 1$; otherwise, it equals 0. After completing the scheduling decisions and notifying the scheduling information via a beacon frame, the coordinator will receive sampled data from the scheduled sources. The scheduling outcome for source i at time t is represented by $p_i(t) \in \{0, 1\}$. For example, if the coordinator does not receive sampled data from scheduled source i at time t, then $p_i(t) = 0$; otherwise, it is set to 1.

Afterwards, the coordinator can determine the AoI for each source node. AoI represents the time elapsed between when data was sampled by source i, denoted as $g_i(t)$, and the current time t. Specifically, the AoI for source i is calculated as [7]

$$A_i(t) = t - g_i(t). \tag{1}$$

The value of $A_i(t)$ increases linearly over time. If a timeslot for source i is not allocated at time t (i.e., $\mu_i(t) = 0$) or the transmission fails from scheduled source i (i.e., $p_i(t) = 0$),

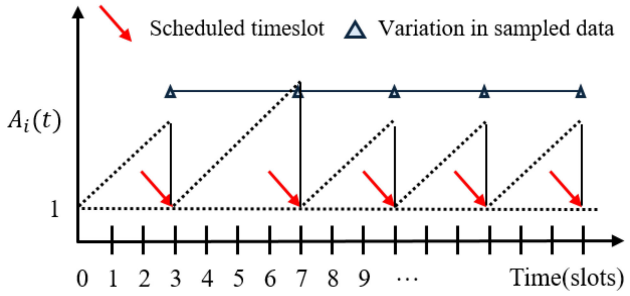


Fig. 3. Motivating scenario: If the information content sampled from source i remains consistent across consecutive instances, should a penalty be imposed on the age of source i at the destination over time?

$A_i(t)$ at the coordinator is updated as

$$A_i(t+1) = \begin{cases} 1, & \text{if } \mu(t) \cdot p_i(t) = 1 \\ A_i(t) + 1, & \text{otherwise.} \end{cases} \tag{2}$$

B. Main Problem: Minimizing Age of Informative Data

The original AoI metric has limitations in quantifying the freshness of information. The fundamental problem lies in defining age. When measuring the age of sampled data, the question arises whether the measure of age should start from the time the information is generated or from the first instance when the information content changes. This problem was first identified in [33]. The authors pose a key question (see Fig. 3): If the information content sampled from source i remains consistent across consecutive instances, should there be a penalty imposed on the age of source i at the destination over time? Subsequent studies stemming from this inquiry propose a novel measurement approach [34], [35], [36], [37], [38], wherein the age begins to increment only when newly sampled data at the source differs from the data stored at the destination (This necessitates the exchange of control signals to detect alterations in information content).

However, this definition is tailored for general networks characterized by intermittent data updates. Especially in WBANs, where source nodes continuously collect physiological data, the information content changes frequently. Hence, this approach may prove ineffective (i.e., potentially leading to increased control overheads) in WBAN-based health monitoring systems. To address the over-sensitivity issue, this work reformulates the original problem of minimizing the age of informative data as minimizing the weighted average AoI, which is defined as

$$\overline{A}' = \lim_{T\to\infty} \frac{1}{T \cdot N} \sum_{t=0}^{T} \sum_{i=0}^{N} \sigma_i(t) \cdot A_i(t) \tag{3}$$

where N is the number of source nodes. σ_i represents the standard deviation of past data observed from source i and is defined as

$$\sigma_i = \sqrt{\frac{\sum_{k=0}^{N} \left(d_i^k - \bar{d}_i\right)^2}{K-1}} \tag{4}$$

where d_i^k and $\bar{d}_i$ are the value of the kth sampled data and the mean of the sampled data observed from source i, respectively. K denotes the number of samples.

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The ultimate goal of this work is to minimize $\overline{A}'$. Pursuing this optimization goal ensures the freshness of information with significant deviations from past observed data, enabling the receiver to promptly and continuously detect emergency situations.

C. Proposed Solution: Semantic-Aware Scheduling

To minimize $\overline{A}'$, the coordinator should make appropriate scheduling decisions for source i at discrete time intervals $t = 0, 1, 2, \ldots$. To develop an appropriate optimization algorithm, the key characteristics of the scheduling problem must be identified: 1) The key decision metric (i.e., standard deviation of the sample), exhibits inconsistency in its value due to the fluctuating nature of the information content over time. For example, historical data collected from a source may show significant deviations in the past, but not at the current stage. This means that a scheduling policy that was optimal in the past may no longer be optimal at the current stage and 2) To identify the informative data, it is important to continuously monitor real-time changes in the information content. This implies that the current scheduling policy should be updated in subsequent beacon periods based on deviations observed from previous beacon intervals.

Hence, given the inherent challenge of predicting future states where the information content varies over time, an approximate solution should be quickly derived (updated) for each beacon period. Hence, it is considered optimal to make decisions greedily (i.e., by making a locally optimal choice), based solely on the current state. The greedy heuristic algorithm is capable of making reasonable, real-time decisions in response to changes in the information content, using a decision metric derived from a combination of the standard deviation and the age metric. Specifically, a combined metric M_i for source i at time t can be calculated as

$$M_i(t) = \sigma_i(t) \cdot A_i(t). \tag{5}$$

Then, the coordinator schedules the source with the highest combined metric M in each timeslot. However, such a greedy decision cannot guarantee near-optimal performance, especially under dynamic channel conditions. As discussed earlier, channel fading induced by frequent body movements impacts transmission reliability. When the coordinator is unable to receive data from a scheduled source due to channel fading, the AoI increases steadily over time. Hence, it is advisable to incorporate channel conditions into the decision metric M. For this, an advanced metric M', which integrates both age and link quality metrics, is computed using a weighted sum, as

$$M'_i(t) = \alpha \cdot \hat{M}_i(t) + (1-\alpha) \cdot \hat{L}_i \tag{6}$$

where α reflects the relative importance of freshness versus link stability. $\hat{M}_i$ and $\hat{L}_i$ are the normalized combined metric M_i for source i and the normalized received signal strength of the last packet received from source i at the coordinator, respectively. It should be noted that, in IEEE 802.15.6, the link quality is primarily conveyed to the MAC layer through the received signal strength indicator [1]. It measures the power level of the received signal at the physical layer (PHY) and provides a direct indication of the link's strength and stability. After each packet reception, the physical layer calculates the received signal strength (i.e., L) and forwards it to the MAC layer.

Here, the reason for normalization is to unify the scale between the two decision metrics. For example, the normalized value of M_i for source i (i.e., $\hat{M}_i$) can be calculated as

$$\hat{M}_i(t) = \frac{M_i(t)}{\max(M_i(t))} \tag{7}$$

where $\mathrm{Max}(M_i(t))$ returns the maximum value of $M_i(t)$ for each source i at time t. The normalized value of L_i for source i (i.e., $\hat{L}_i$) is also calculated in the same way as follows:

$$\hat{L}_i = \frac{L_i}{\max(L_i)}. \tag{8}$$

For each slot t, the coordinator schedules the source with the highest $M'(t)$. This ensures that the scheduled source not only maintains a reliable connection but also delivers informative data, ultimately minimizing $\overline{A}'$. However, to achieve such an optimal outcome, the challenge of determining the weight α must be tackled.

The challenge of determining the weight α involves balancing choices that are safe but have low expected returns against those that are risky but have high expected returns. For example, scheduling a source with the largest deviation from historical data and the greatest AoI may yield the highest return (i.e., in terms of minimizing $\overline{A}'$). However, it may be the worst decision if the scheduled source fails to transmit due to dynamic channel conditions. This is because the AoI of the scheduled source is not reset due to transmission failures and thus increases steadily over time. On the other hand, scheduling a source with small variation in data and relatively low AoI may not be expected to yield the best outcome, but it can be the best decision in dynamic situations if the scheduled source has good channel conditions and successfully transmits the sampled data.

To solve such a balancing problem, this article introduces a self-adaptive greedy algorithm. The previous work [57] verified that it is possible to infer future channel conditions using personalized movement patterns in WBANs. Building on this work, the proposed technique learns long-term channel statistics and adjusts the weights α based on future channel conditions such that $\overline{A}'$ is minimized. In other words, the coordinator extracts personalized channel characteristics using deep reinforcement learning (DRL) to identify the robustness of channel conditions. Based on the learning results, the coordinator can assign a higher weight to $\hat{M}$ under stable channel conditions, while assigning a higher weight to $\hat{L}$ under unstable channel conditions. This approach can provide a flexible tradeoff between conflicting decision metrics, ultimately minimizing $\overline{A}'$. Further details are given in Section V.

IV. Modeling of Upper Layer

To facilitate semantic-aware communication, an upper layer positioned between the MAC and APP layers is introduced in this article. This section provides a high-level overview

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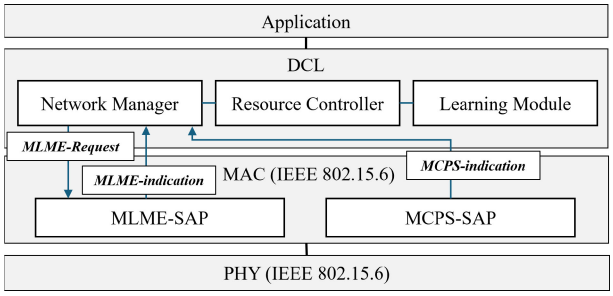


Fig. 4. High-level view of the protocol architecture.

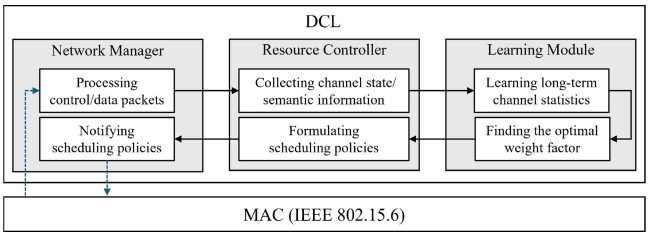


Fig. 5. Overview of the DCL components.

of the architectural structure of the upper layer, referred to as the dynamic control layer (DCL), highlighting its key components and interfaces. The structure adheres to the IEEE 802.15.6 standard. The following subsections explore the motivation behind designing the upper layer and offer detailed descriptions of each component.

A. Motivation for Designing the DCL

The core components that enable semantic-aware scheduling include: 1) observing deviations in data received from each source and 2) updating the AoI for each source. However, these tasks align more closely with the responsibilities of the APP layer and therefore lie outside the purview of the MAC layer. In addition, the self-adaptive mechanism requires cooperation with the MAC layer to learn long-term channel statistics (refer to the following subsections). This increases the complexity of the MAC layer and reduces its compatibility with other systems. To address this issue, the DCL was designed to collect network information and self-adapt the scheduling policy. This new design integrates network configuration, resource allocation, and adaptive learning techniques. Further details about the DCL are provided in the following subsections.

B. Protocol Architecture

Fig. 4 shows a high-level view of the protocol architecture on the coordinator side. The extended protocol architecture is based on the IEEE 802.15.6 standard and consists of four layers: PHY, MAC, DCL, and APP. The PHY and MAC layers are based on the IEEE 802.15.6 standard, and the MAC layer is divided into two types of service access points (SAPs): 1) the MAC sublayer management entity (MLME) and 2) the MAC common part sublayer (MCPS). Specifically, the MLME-SAP manages network access, synchronization, and parameter configuration, while the MCPS-SAP handles data transmission and confirmation. Further details about this concept can be found in [1]. The DCL serves as an interface that connects the upper and lower layers (the corresponding service primitives are defined in the following section). Moreover, the DCL works closely with the MLME-SAP in the MAC layer to effectively manage and control the network, and to establish resource allocation policies. The APP layer gathers and consolidates physiological data, which it then transmits to a remote monitoring server via an external network.

C. Definition of Service Primitives

As shown in Fig. 4, service primitives are defined for communication between the upper layer and the MAC layer. These service primitives serve as essential interfaces that facilitate communication and control between the layers. Specifically, two types of service primitives are used.

1) Network Management Primitive: To initiate the transmission of network management frames, the network manager in the DCL sends control signals to the MLME-SAP in the form of an MLME-request. Upon receiving a control signal (i.e., an MLME-request), the MLME-SAP constructs a network management frame (e.g., a beacon frame) and sends it either to a specific destination or broadcasts it to the entire network. Concurrently, the MLME-SAP forwards network management frames received from each source to the network manager in the form of an MLME-indication.

2) Data Transfer Primitive: To initiate the transmission of sampled data, the application at the source side directly sends a control signal to the MCPS-SAP in the form of an MCPS-request. Upon receiving a control signal, the MCPS-SAP constructs a data frame and transmits it to the coordinator. Meanwhile, the MCPS-SAP at the coordinator side notifies the network manager of incoming data frames in the form of an MCPS-indication.

D. Overview of the DCL Components

Fig. 5 presents a high-level overview of the interactions and responsibilities of the DCL components, which include the network manager, resource controller, and learning module. Each component is briefly described as follows.

1) Network Manager: The network manager serves as an interface for communication with the MAC layer and other DCL components. Specifically, the network manager receives data and control messages from the MAC layer and provides the required information to the resource controller for establishing scheduling policies. In addition, it handles the initiation of association procedures while coordinating the periodic broadcasting of beacon frames to maintain synchronization and notify scheduling information.

2) Resource Controller: The resource controller is responsible for collecting channel state and semantic information from the network and formulating scheduling policies. To enable this, the network manager forwards the data (including the channel state information) received from scheduled source i to the resource controller. Then, the resource controller updates A_i for source i at time $t = 0, 1, 2, \ldots$ [see (2)]. Additionally, it calculates the standard deviation σ_i from

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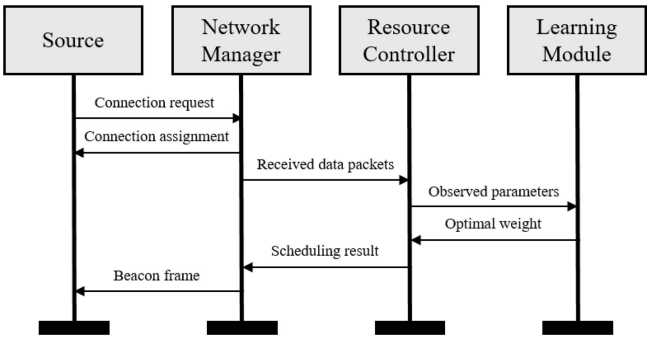


Fig. 6. Sequence chart illustrating the basic operation.

historical data at source i over N periods. Further details on these operations are provided in the next section.

3) Learning Module: The learning module provides machine learning support, enabling the DCL to train models and use them for making inferences. Specifically, the resource controller predicts future channel conditions by extracting personalized movement patterns of WBAN users with the help of the learning module and uses the learning results to derive scheduling policies. Further details are given in the next section.

V. Semantic-Aware Scheduling Method

A. Basic Operation

This section outlines the coordinated process chain within the DCL. Detailed steps are provided in Algorithm 1 and the subsequent explanations.

1) Initialization: In IEEE 802.15.6, network initialization between source nodes and the coordinator involves an association procedure with connection request and connection assignment frames. As shown in Fig. 6, a source node sends a connection request frame to the coordinator, which is processed as an MLME-indication service primitive via the MLME-SAP to the network manager within the DCL. If the network manager approves the request, it sends an association grant back to the MLME-SAP as an MLME-request service primitive. The MLME-SAP then generates and forwards a connection assignment frame to the source node. Once the source node receives this frame, it is officially associated with the coordinator and can start participating in the network.

2) Timeslot Scheduling: Once the association is established, the resource controller within the DCL performs the initial timeslot scheduling. The initial weight α is set to 0.5, and the initial values of σ_i, A_i, and L_i for each source i are all set to 1 (these values will be updated over time). For each slot t, the resource controller calculates $M_i'(t)$ [see (6)] for each source i and schedules the source with the highest $M'(t)$. After finishing the initial scheduling, the resource controller forwards the scheduling information to the network manager.

3) Parameter Update: The network manager encapsulates the scheduling result in a beacon frame and initiates the transmission by sending an MLME-request to the MLME-SAP. After sending the beacon frame, the network manager forwards the data packet received from scheduled source i, along with the received signal strength (i.e., L_i), to the resource controller. Upon receiving the data packets from the network manager, the resource controller updates A_i [see (2)] and σ_i [see (4)] for each scheduled source i. At the end of the current beacon period, the resource controller forwards the observed parameters (i.e., A_i, σ_i, and L_i) to the learning module.

Algorithm 1 Weights Assignment Algorithm for Multimodal Classifier

Resource Controller:
1: Initialize parameters: $\alpha \leftarrow 0.5$, $\sigma_i \leftarrow 1$, $A_i \leftarrow 1$, $L_i \leftarrow 1$ for each source i.
2: Invoke **Slot_Allocation**(α) for initial scheduling.
3: Forward the scheduling result to the Network Manager.
4: **Upon** receiving the updated α:
5: Invoke **Slot_Allocation**(α) to derive a new scheduling result.
6: Forward the scheduling result to the Network Manager.
Network Manager:
7: **Upon** receiving the scheduling result:
8: Notify the scheduling information via a beacon frame.
9: **Upon** receiving a data packet from scheduled source i:
10: Update L_i, A_i, and σ_i using the received packet.
11: Forward the updated parameters to the Learning Module.
Learning Module:
12: **Upon** receiving the updated parameters L_i, A_i, and σ_i:
13: Optimize the weight α for the next beacon period.
14: Forward the updated α to the Resource Controller.

4) Weight Adjustment: Upon receiving the observed parameters from the resource controller, the learning module determines the optimal weight α for the upcoming beacon period. Specifically, the DRL engine evaluates the tradeoff between a conservative decision with lower expected returns and a risky decision with higher expected returns, focusing on minimizing $\overline{A}'$. After finding the optimal weight α, the learning module forwards it to the resource controller. Upon receiving the updated weight, the resource controller derives a new scheduling result for the next beacon period and forwards it to the network manager. The subsequent procedures are the same as those described in (3) and (4).

B. Scheme Details: Timeslot Scheduling

In WBANs, time is divided into frames (i.e., superframes), and the frames are further divided into timeslots. Each source can be scheduled in any timeslot within the frame, and the value of A_i resets to 1 if the transmission of scheduled source i succeeds; otherwise, A_i will increase over time. Since a WBAN operates based on beacon mode, the coordinator should make scheduling decisions on a frame-by-frame basis, rather than on a slot-by-slot basis. It is assumed that the beacon interval is equal to the superframe length.

Timeslot scheduling is performed by the resource controller at the end of the current beacon period, just before the start of the next beacon period (see Fig. 2). After sending the first beacon frame (which includes the initial scheduling information), the resource controller updates A_i, σ_i, and L_i for each scheduled source i based on the received packets. At

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Algorithm 2 Self-Adaptive Greedy Scheduling

Initialization:

P_{max} ← Maximum number of timeslots (within the superframe)

$V[P_{max}] \leftarrow \varnothing$ // Resource allocation vector (timeslot grant queue)

Function: Slot_Allocation (weight α)

1: **for** $t \leftarrow 0, t < V_{max}$ **do**
2: **for** all sources i **do**
3: $M_i(t) \leftarrow \sigma_i(t) \cdot A_i(t)$ // Eq. (5)
4: $\hat{M}_i(t) \leftarrow \frac{M_i(t)}{\max(M_i(t))}$ // Eq. (7)
5: $\hat{L}_i \leftarrow \frac{L_i}{\max(L_i)}$ // Eq. (8)
6: $M'_i(t) \leftarrow \alpha \cdot \hat{M}_i(t) + (1-\alpha) \cdot \hat{L}_i$ // Eq. (6)
7: $A_i(t+1) \leftarrow A_i(t) + 1$ // Eq. (2)
8: **end for**
9: $k \leftarrow \arg\max(M'_i(t))$ // Return the index of the source with the highest $M'(t)$
10: $V[t] \leftarrow k$ // Allocate a timeslot to the source with the highest $M'(t)$
11: /* If source i is scheduled at slot t */
12: **if** $k = i$ **then**
13: $A_i \leftarrow 1$ // Reset the AoI of scheduled source i
14: **end if**
15: $t \leftarrow t + 1$
16: **end for**

the end of the current beacon period, the resource controller forwards the observed parameters (i.e., A_i, σ_i, and L_i) to the learning module. As described above, the learning module returns the optimal weight α for the upcoming timeslot scheduling. After receiving the updated weight, the resource controller initiates timeslot scheduling for the next beacon period.

Algorithm 2 outlines the operation of the proposed timeslot scheduling algorithm. For each slot t (within the superframe), the resource controller derives the decision metric $M'_i(t)$ for source i using the weighted sum equation (lines 3–6). The source with the highest M' is then scheduled in slot t (lines 9 and 10). Once source i is scheduled in the current slot t, A_i is reset to 1, regardless of the actual transmission result (lines 11–14). This ensures that all sources are competing fairly in the next slot $t+1$ and this reset does not affect actual performance.

Time Complexity: Algorithm 2 is a polynomial-time algorithm. The time complexity can be analyzed based on its nested loops and operations. The outer loop iterates over each timeslot, with a maximum of $P_{\max}$ iterations. Within this loop, there is an inner loop that processes all sources N. For each source, several operations are performed, including calculating the combined metric $M_i(t)$, normalizing both $M_i(t)$ and L_i, and computing the weighted sum $M'_i(t)$, all of which take constant time $O(1)$. After processing all sources, the algorithm selects the source with the maximum $M'_i(t)$, which requires $O(N)$ time. As these operations are repeated for each timeslot, the overall time complexity of the algorithm is $O(P_{\max} \times N)$, where $P_{\max}$ is the number of timeslots and N is the number of sources.

C. Scheme Details: Weight Adjustment

The basic idea for determining the weight α is as follows: if the channel is predicted to be dynamic in the next beacon period, the weight of $\hat{L}_i$ is increased. Conversely, if the network is expected to remain stable, the weight of $\hat{M}_i$ is increased. In other words, the weights are determined based on the predicted channel conditions in the near future (i.e., the next beacon period). As discussed earlier, channel conditions in WBANs are directly influenced by body movements. Therefore, predicting future channel conditions requires anticipating future body movements. Fortunately, human movements tend to follow personalized patterns based on individual lifestyles, and by analyzing the combination of signal strengths from sensors attached to different body parts, both the current posture and the channel conditions resulting from subsequent movements can be inferred.

To address this weight determination problem, Q-learning is employed. By leveraging Q-learning to recognize and predict user movement patterns, the relative weights between the two conflicting decision criteria can be adjusted based on long-term channel statistics in personalized WBANs. This approach enables the system to adapt to the unique movement patterns of individuals and optimize weight allocation in response to future channel conditions. Specifically, future channel conditions are predicted based on personalized body movement patterns by analyzing the correlation between the received signal strengths collected from each beacon interval. To implement Q-learning, the system is modeled based on key components as follows.

1) State Space: The state space s at each beacon interval is defined as

$$s = \left[\{\overline{A}_1, \sigma_1, \overline{L}_1\}, \{\overline{A}_2, \sigma_2, \overline{L}_2\}, \ldots, \{\overline{A}_N, \sigma_N, \overline{L}_N\}\right] \tag{9}$$

where $\overline{A}_1$ and $\overline{L}_1$ are the average values of A_1 and L_1 for source number 1 within the last beacon period. N is the number of source nodes.

2) Action Space: The action space a consists of a vector of weights. These weights decrease incrementally from 0.9 to 0.1, such that

$$a = [0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1]. \tag{10}$$

Each weight in a corresponds to a proportion of the relative importance between two conflicting decision criteria (i.e., $\hat{M}_i$ and $\hat{L}_i$).

3) Reward Function: The reward function $r(s, a)$ is formulated to reflect network performance

$$r(s,a) = -\frac{1}{T \cdot N} \sum_{t=0}^{T} \sum_{i=0}^{N} \sigma_i(t) \cdot A_i(t) \tag{11}$$

where T is the number of timeslots within the superframe. Based on these components, the Q-learning algorithm updates the Q-values for state-action pairs as following:

$$Q(s,a) \leftarrow Q(s,a) + \eta\left[r(s,a) + \gamma \max_{a'} Q(s',a') - Q(s,a)\right] \tag{12}$$

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Algorithm 3 DQN Training for Adjusting the Weight α

Initialize Q-network with random weights θ.
Initialize target network with weights θ^- where $\theta^- = \theta$.
Initialize replay buffer $\mathcal{D}$.
for each episode **do**
  Initialize environment and get initial state s_0:

$$s_0 = [\{\overline{A}_1, \sigma_1, \overline{L}_1\}, \{\overline{A}_2, \sigma_2, \overline{L}_2\}, ..., \{\overline{A}_N, \sigma_N, \overline{L}_N\}]$$

  for each time step t **do**
    With probability ϵ, select a random action a_t from the action space:

$$a_t = [0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1]$$

    Otherwise, select action $a_t = \arg\max_a Q(s_t, a; \theta)$.
    Execute action a_t in the environment.
    Observe reward r_t and next state s_{t+1}.
    Store transition (s_t, a_t, r_t, s_{t+1}) in replay buffer $\mathcal{D}$.
    Sample random minibatch of transitions (s_j, a_j, r_j, s_{j+1}) from $\mathcal{D}$.
    Compute target for each minibatch sample:

$$y_j = \begin{cases} r_j & // \text{ if episode ends at step } j+1, \\ r_j + \gamma \max_{a'} Q(s_{j+1}, a'; \theta^-) & // \text{ otherwise.} \end{cases}$$

    Perform gradient descent on loss function:

$$L(\theta) = \frac{1}{N} \sum_j \left(y_j - Q(s_j, a_j; \theta)\right)^2$$

    Periodically update target network weights θ^-:
  end for
end for

where η is the learning rate and γ is the discount factor. $r(s, a)$ is the reward received after taking action a in state s. s' is the next state resulting from action a. $\max_{a'} Q(s', a')$ is the maximum Q-value for the next state s' over all possible actions a'.

By applying Q-learning, the learning module can obtain the appropriate weight α for the next beacon period. However, as the state and action space grow larger and more complex, traditional Q-learning methods struggle with the curse of dimensionality, making it increasingly difficult to effectively estimate the Q-values for each state-action pair. This limitation calls for a more scalable approach, where function approximation methods are employed to generalize Q-value estimations across the state-action space. In such scenarios, a deep Q-network (DQN) becomes essential. By leveraging deep learning techniques, DQN can approximate the Q-values using a neural network, enabling the learning module to handle high-dimensional state spaces and make more informed decisions in complex environments.

Algorithm 3 describes the training procedure for adjusting the weight α in the learning module. Initially, a Q-network with random weights and a target network with the same weights are set up, along with a replay buffer. For each episode, the environment is initialized with the state vector s_0. At each time step, an action is selected either randomly (with probability ϵ) or by using the Q-network to choose the action with the highest predicted reward. The selected action is executed, yielding a reward and the next state, which are stored in the replay buffer. A minibatch of experiences is sampled from the buffer to compute the target values, based on whether the episode ends or continues. The loss function is then calculated as the mean squared error between the target and predicted Q-values, and gradient descent is applied to minimize this loss. Periodically, the weights of the target network are updated to match those of the Q-network, ensuring stable learning.

VI. Performance Evaluation

A. Testbed Setup: Overview of the BANSIM

To verify the performance of the proposed scheduling scheme, the body area network simulator (BANSIM) [58] was used. BANSIM is a Python-based DRL-enabled discrete-event network simulator for WBANs, designed to simulate network models, run custom protocol stacks, and transmit packets over wireless links. It provides a lightweight environment for testing with minimal network functions, and its key advantage is that it does not require a framework or middleware for DRL integration.

As illustrated in Fig. 7, the architecture of BANSIM consists of four network models (i.e., APP, service-specific convergence sublayer (SSCS), MAC, and PHY) and four supplementary components (i.e., DQN trainer, channel model, mobility model, and packet model), each fulfilling specific roles. Protocols are organized using a class model, allowing direct communication between adjacent layers through interface methods. Specifically, the SSCS layer was customized to implement the proposed control layer (i.e., DCL). BANSIM supports the creation of a WBAN environment with one coordinator and N sensor nodes, connected in an one-hop star topology. The simulator also supports TDMA for channel access, with customizable beacon structures and periods.

1) Channel Model: In BANSIM, the channel model facilitates communication between the sender and receiver, allowing users to implement various propagation loss models. It supports two main models: 1) the Friis free-space model and 2) a BAN-specific path-loss model. The Friis model predicts reception power based on transmission power, antenna gain, wavelength, and distance, assuming a line-of-sight (LOS) condition. However, due to frequent body movements in WBANs, non-LOS (NLOS) scenarios often occur, requiring consideration of body-shadowing effects for realistic channel modeling.

The WBAN-specific path-loss model is derived from real-WBAN testbed data [1] and provides a better fit for predicting reception power in WBAN environments. It is expressed as

$$PL(d) = PL(d_0) + 10n \log_{10}(d/d_0) + S \tag{13}$$

where d is the distance between sender and receiver, $PL(d_0)$ is the path loss at reference distance d_0, n is the path-loss exponent, and S is the body-shadowing factor. When LOS is

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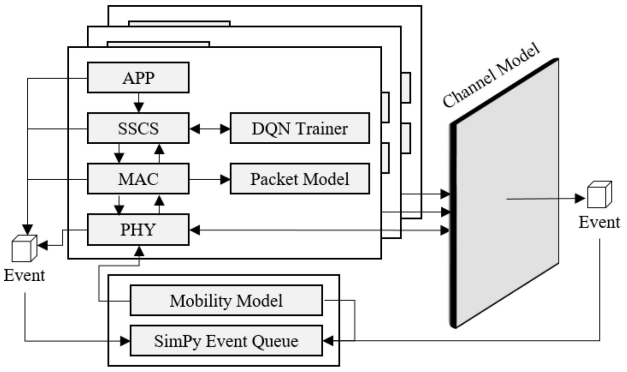


Fig. 7. Overview of the BANSIM architecture.

guaranteed, the shadowing factor is omitted. LOS or NLOS conditions are determined using the mobility model.

2) Mobility Model: BANSIM offers three human mobility models: standing, sitting, and walking. These models are represented by 15 position vectors, each corresponding to a body part (e.g., torso, knee, and wrist), which move or rotate periodically. Based on these movements, the body-shadowing factor and path loss are calculated. Reception power is determined by transmission power, path loss, and noise. If the reception power falls below the receiver's sensitivity, the packet is discarded.

B. Simulation Settings

The simulation environment consists of one coordinator and eight body sensors, forming an one-hop star topology. Both the coordinator and sensor nodes adhere to the IEEE 802.15.6 standard and operate in the 2.4 GHz frequency band. Communication between the coordinator and sensor nodes uses quadrature phase shift keying (QPSK) modulation, with a data rate of 250 Kb/s. The transmission power is set to 0 dBm, and the reception sensitivity is -82 dBm. The network employs TDMA for channel access, and communication follows a beacon-enabled mode with superframes. Each beacon period is 250 ms, and the acknowledgment policy follows immediate acknowledgment (I-ACK). The slot size is configured to 8 ms, and the traffic model assumes a full buffer model to ensure continuous data transmission. The specific simulation parameters used in the tests, including the hyperparameters for DQN training, are provided in Tables I and II, respectively. The simulation was run for 3600 s and conducted 100 times, with a 95% confidence interval.

1) Mobility Settings: To simulate realistic human mobility, we configure two types of test networks.

1) In Test Network A, the WBAN user exhibits completely random movements, with unpredictable positions and rotations of the same body parts. This randomness leads to dynamic variations in the body-shadowing factor and path loss.
2) In Test Network B, the WBAN user follows personalized movement patterns by periodically switching between standing, sitting, and walking models. These models are represented by specific body parts (e.g., left elbow, right wrist, left knee) listed in Table III. The movements and rotations of these body parts are used to simulate realistic body movement and calculate the body-shadowing factor, which impacts path loss.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Frequency band	2.4 GHz
Modulation	QPSK
Date rate	250 Kbps
Transmission power	0 dBm
Reception sensitivity	-82 dBm
Channel access	TDMA
Access mode	Beacon with superframes
Beacon period	250 ms
Ack policy	I-ACK
Slot size	8 ms
Traffic model	Full buffer
Mobility model	Personalized or random

TABLE II
HYPERPARAMETERS

Parameter	Value
Neural network model	Sequential
Number of layers	2
Number of neurons per layer	24
Input dimension	2
Activation function	ReLU
Optimizer	RMSProp
Exploration probability (ϵ)	0.9
Exploration rate decay	0.01
Minimum replay memory size	1000
Replay memory size	100000
Update frequency of target-network	5
Reward discount factor	0.99
Batch size	64

2) Application Settings: Each sensor is attached to a different body part and generates physiological data based on the application type. The full buffer model is used, meaning the sensors always have data ready to send. At each scheduled slot, the sensor transmits the most recent data in its buffer. The slot size and sensor's sampling period are synchronized to 8 ms, allowing each sensor to collect a new sample at the start of every slot. To account for variations in information content, the data range is randomly set between 1 and 120. Further details on the application settings can be found in Table III.

C. Simulation Objective

Based on the simulation settings, three key performance indicators (KPIs) were evaluated: average AoI, weighted average AoI, and system throughput. The average AoI was analyzed to demonstrate that the proposed scheme effectively maintains the freshness of data sampled from each source. The weighted average AoI was used to verify that the proposed scheme ensures the freshness of informative data, particularly

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TABLE III
NODE (APPLICATION) SETTINGS

NID	Packet size	Sampling period	Data range	Content change interval	Node position
0	50 - 250 bytes	8 ms	Random (1, 50)	50 - 250 ms	Left elbow
1	50 - 250 bytes	8 ms	Random (1, 60)	50 - 250 ms	Left wrist
2	50 - 250 bytes	8 ms	Random (1, 70)	50 - 250 ms	Right elbow
3	50 - 250 bytes	8 ms	Random (1, 80)	50 - 250 ms	Right wrist
4	50 - 250 bytes	8 ms	Random (1, 90)	50 - 250 ms	Left knee
5	50 - 250 bytes	8 ms	Random (1, 100)	50 - 250 ms	Left ankle
6	50 - 250 bytes	8 ms	Random (1, 110)	50 - 250 ms	Right knee
7	50 - 250 bytes	8 ms	Random (1, 120)	50 - 250 ms	Right ankle

when it significantly deviates from historical data. By analyzing system throughput, it is possible to demonstrate that the proposed scheme effectively minimizes packet loss caused by body shadowing, reducing resource waste and lowering the overall average AoI.

Through the analysis of these KPIs, the key facts discussed in the literature review can be verified.

1) The original AoI metric [15] increases linearly over time. This means that even if two consecutive samples contain the same information, unnecessary penalties are imposed. As a result, it fails to minimize the average and weighted average AoI, particularly when ensuring the freshness of informative data in WBAN-based health monitoring systems.
2) The latest semantic-aware age metric (i.e., AoII) [37] is overly sensitive to information content, often imposing penalties for minor variations. As a result, it tends to behave like the original AoI metric, with penalties increasing over time.
3) The proposed combined metric M [see (5)] can ensure the freshness of informative data with substantial deviations from past data, enabling more prompt and continuous detection of emergency situations in WBAN-based health monitoring systems.
4) The proposed advanced metric M' that integrates both age and link quality metrics [see (6)] can improve system performance across the three KPIs in WBAN-based health monitoring systems.

To ensure a fair comparison among the four metrics, the same scheduling policy (i.e., greedy scheduling) was applied uniformly during the simulation.

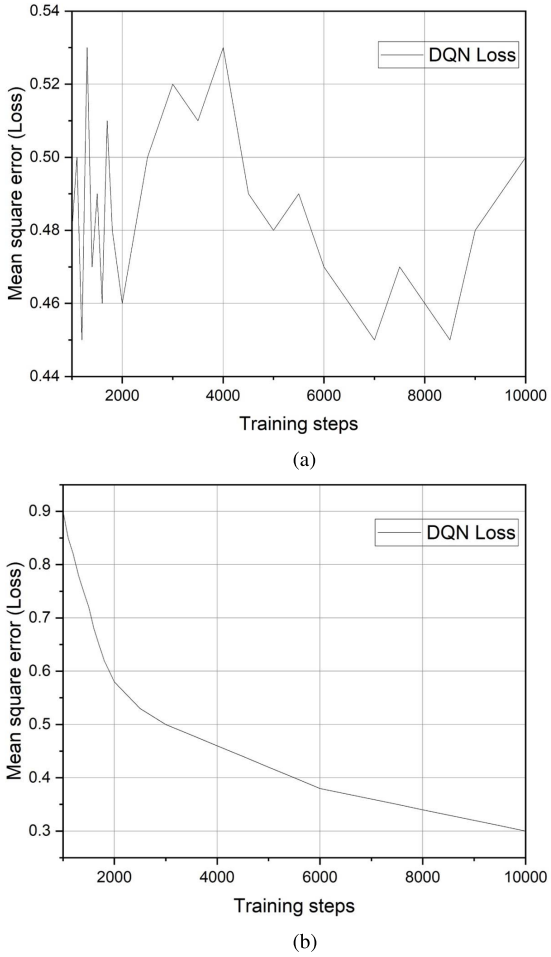


Fig. 8. Convergence of the DQN loss function. (a) Test network A. (b) Test network B.

D. Analysis of Loss Convergence

The convergence of the DQN loss function requires the presence of consistent relationships and patterns between states, actions, and subsequent states. When personalized movement patterns exist (i.e., Test Network B), the link quality variations follow predictable transitions, enabling stable Q-value predictions and gradual loss reduction. However, in scenarios with random movements (i.e., Test Network A), the lack of predictable patterns leads to erratic Q-value updates, making it difficult for the loss to converge. Therefore, achieving convergence depends on the system's ability to maintain predictable state-action transitions. In this regard, as shown in Fig. 8(a), the DQN loss fluctuates consistently without converging. On the other hand, as shown in Fig. 8(b), the DQN loss gradually converges over the training steps.

E. Analysis of KPIs: Average AoI

The system's average AoI represents the average AoI of the data received from each source from the destination's perspective. It captures the elapsed time between when data are generated and when it is received, making it a fundamental performance metric for measuring data freshness. In other

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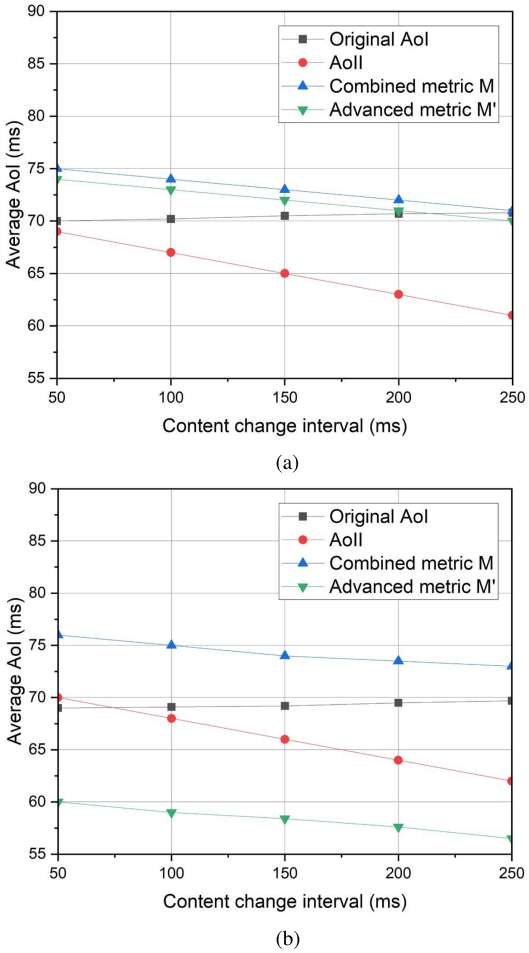


Fig. 9. Performance comparison: average AoI (varying content change interval). (a) Test network A. (b) Test network B.

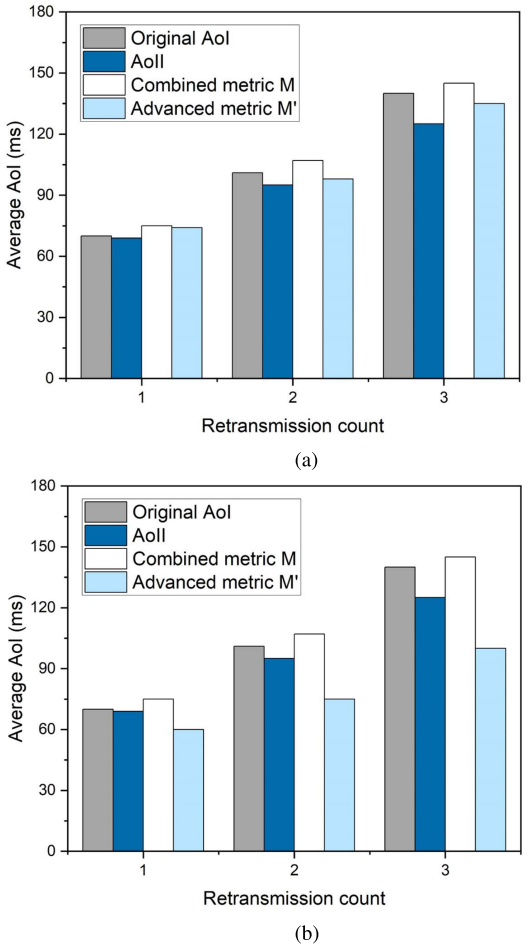


Fig. 10. Performance comparison: average AoI (varying retransmission count). (a) Test network A. (b) Test network B.

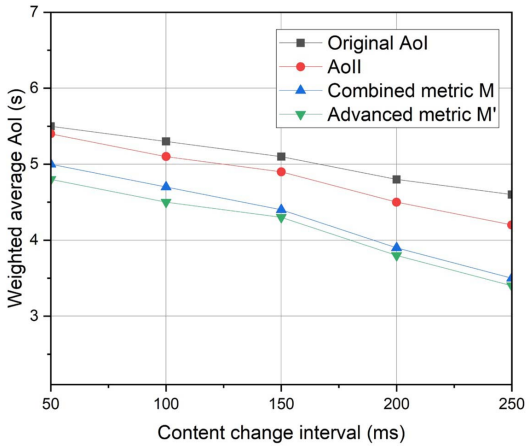
words, a lower system average AoI indicates fresher data collected from sensors within that system. This section first examines how average AoI changes according to the update interval of the information content generated by each source. A shorter update interval results in more frequent changes in information content, while a longer update interval implies fewer changes. By dynamically adjusting the content change interval, the average AoI was measured, and the advantages and disadvantages of each age-related metric were discussed.

In both the original AoI metric and AoII metric cases, different perspectives are considered for AoI calculation. The AoII metric increases the AoI of each source when the information content at the source differs from that at the destination (ignoring signal overhead). However, as shown in Fig. 9(a) and (b), the AoII metric performs similarly to the original AoI metric when the content change interval is short. This is because, in health monitoring systems, data are frequently updated, and the information content between the source and destination is likely to differ. Thus, as noted in the literature review, AoII ends up penalizing the age using the same mechanism as the original AoI metric, leading to similar performance between the two. The combined metric M shows slightly lower performance compared to both the original AoI and AoII metrics. This is because the scheduler prioritizes sources with high AoI, weighted by the standard deviation of their data at each slot, focusing on collecting more informative data. As a result, sources with high data variance may be scheduled first, even if their AoI is relatively high.

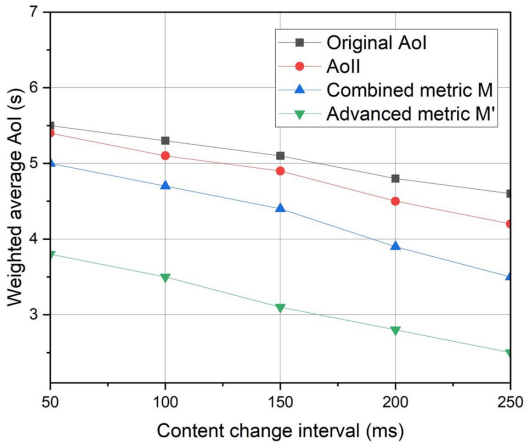
However, if a scheduled source experiences continuous data loss due to body movement, the AoI at the destination will keep increasing (due to transmission failures from body shadowing and the inability to reset the AoI of sources not scheduled). The scheduler will continue attempting to schedule that source, resulting in repeated transmission failures, wasting time resources, and increasing the AoI of other sources as well. This ultimately raises the system's average AoI. As discussed in the literature review, scheduling decisions based solely on current or future AoI cannot account for the increase in average AoI caused by transmission failures. Therefore, a flexible tradeoff between transmission failures and the increase in current or future AoI must be provided to effectively minimize the system's average AoI.

From this perspective, the reason why the original AoI metric, AoII metric, and combined metric M show similar performance in terms of average AoI is that they do not integrate channel conditions into the decision metric. As a result, none of these metrics can prevent the AoI increase caused by transmission failures, leading to higher average

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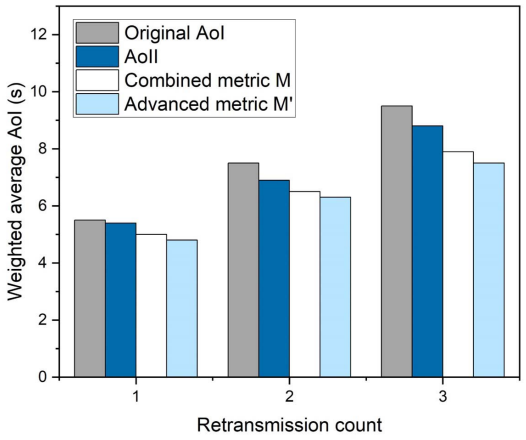


(a)

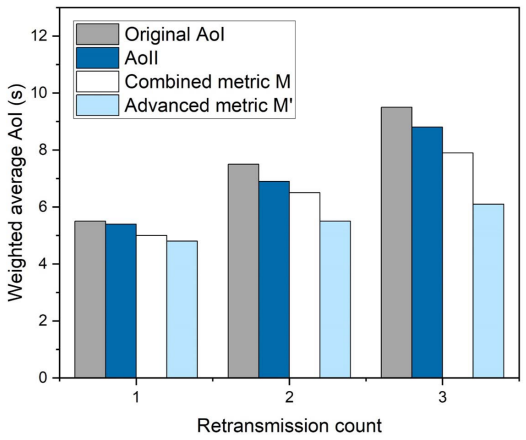


(b)

Fig. 11. Performance comparison: weighted average AoI (varying content change interval). (a) Test network A. (b) Test network B.



(a)



(b)

Fig. 12. Performance comparison: weighted average AoI (varying retransmission count). (a) Test network A. (b) Test network B.

AoI. On the other hand, the advanced metric M' integrates channel conditions into the decision metric and determines the optimal weight between the age and link quality metrics based on future channel conditions, thus finding a flexible tradeoff between the two conflicting metrics.

As shown in Fig. 9(a), in Test Network A, where random movements are present, predicting future channel conditions is difficult, making it hard to prevent packet loss due to body shadowing. Therefore, all metrics perform similarly, but the advanced metric M' achieves the lowest average AoI by excluding sources with poor current channel conditions from scheduling, even if their AoI is high. In contrast, as shown in Fig. 9(b), in Test Network B with personalized movement patterns, the advanced metric M' shows particularly superior performance. This is because it can predict changes in channel quality based on the channel statistics of each source and the user's future body posture, effectively addressing the issue.

Changes in average AoI according to the number of retransmissions were also captured to analyze the impact of transmission failures on average AoI in more depth. (*Note:* The default retransmission count defined by the WBAN standard is 3.) Clearly, the more times the same sample is retransmitted due to transmission failures, the more the time increases from when the data is generated to when it reaches the destination. Based on this, the effect of transmission failures on average AoI was analyzed.

As shown in Fig. 10(a), in Test Network A with random movements, predicting future channel conditions is difficult, resulting in high packet loss due to body shadowing when scheduling policies only consider AoI. Although the advanced metric M' integrates dynamic channel conditions into the scheduling decisions, it cannot effectively suppress AoI increases caused by packet loss due to unpredictable body movements. As a result, all metrics show similar performance. However, as shown in Fig. 10(b), in Test Network B with personalized movement patterns, the advanced metric M' performs best. This is because it predicts future changes in channel quality based on the channel statistics of each source's body posture, effectively suppressing AoI increases caused by transmission failures.

F. Analysis of KPIs: Weighted Average AoI

In this section, changes in weighted average AoI were captured according to the update interval of information content generated by each source and the number of retransmissions.

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The weighted average AoI is an indicator that weights the average AoI of data received from each source at the destination by the data's deviation, serving as a performance metric for measuring the freshness of emergency data. In other words, the lower the system's weighted average AoI, the fresher the informative data collected from sensors remains within that system.

The weighted average AoI gradually increases if sources are not scheduled in each slot, just like the calculation of average AoI. While scheduling the source with the highest AoI in each slot can effectively minimize average AoI, it cannot effectively minimize weighted average AoI if the received data's information content is not significantly different from the previously collected data. Therefore, the scheduler must consider not only sources with high AoI but also those with high data deviation in each slot, as this deviation is weighted against AoI. Thus, the scheduler should prioritize sources with both high standard deviation and AoI in each slot.

From this perspective, as seen in Fig. 11(a) and (b), the original AoI metric and AoII metric cannot achieve the minimization of weighted average AoI (i.e., they struggle to maintain the freshness of informative data). Specifically, while the AoII metric increases the AoI of each source when the data collected at the source differs from the data at the destination, it shows similar performance to the original AoI metric when the update interval is short. On the other hand, the combined metric M and advanced metric M' assign resources to sources that are producing relatively more meaningful data by weighting the deviation of previously collected data into the current AoI during scheduling decisions. As a result, these metrics perform better than the original AoI metric and AoII metric.

Furthermore, as discussed in the previous section, transmission failures due to body shadowing lead to an increase in average AoI, which also directly impacts the weighted average AoI. As shown in Fig. 11(a), in Test Network A with random movements, it is difficult to predict future channel conditions, making it hard to prevent packet loss due to body shadowing. Therefore, all metrics show similar performance. However, the advanced metric M' achieves the lowest weighted average AoI by excluding sources with poor current channel conditions from being scheduled, even if their AoI is high. As seen in Fig. 11(b), in Test Network B with personalized movement patterns, the advanced metric M' shows significantly better performance. This is because it can predict changes in channel quality based on the channel statistics of each source and the user's future body posture, effectively addressing the issue.

Changes in weighted average AoI were also captured according to the number of retransmissions. As discussed earlier, the more times the same sample is retransmitted due to transmission failures, the more the time from data generation to its arrival at the destination increases. As shown in Fig. 12(a), in Test Network A with random movements, it is difficult to predict future channel conditions, and thus scheduling policies that only consider AoI experience significant packet loss due to body shadowing. Despite the advanced metric M' incorporating dynamic channel conditions into scheduling decisions, it fails to effectively mitigate the AoI increase resulting from packet loss due to unpredictable body movements. Consequently, all metrics exhibit comparable performance. As shown in Fig. 12(b), in Test Network B, where personalized movement patterns are present, the advanced metric M' outperforms the others. Its ability to predict future channel quality fluctuations based on each source's channel statistics and the user's body posture allows it to effectively mitigate the AoI increase caused by transmission failures.

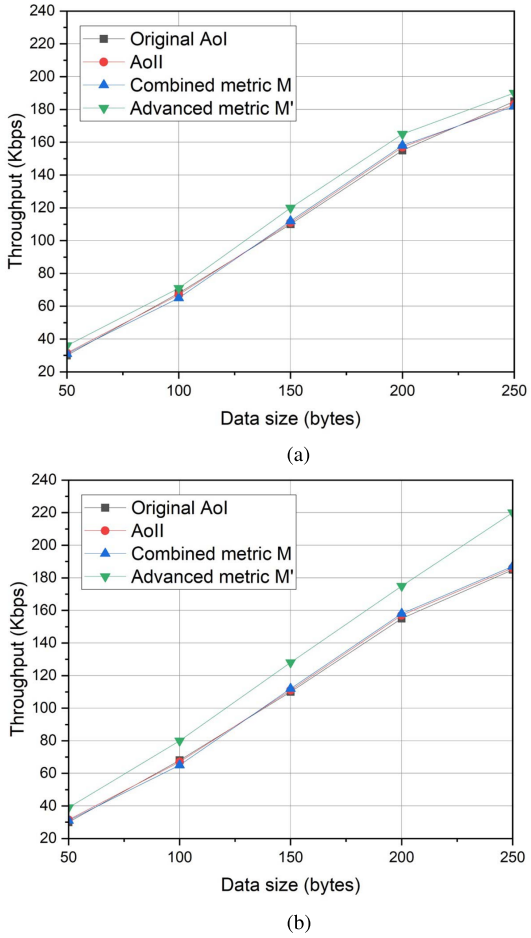


Fig. 13. Performance comparison: throughput (varying data size). (a) Test network A. (b) Test network B.

G. Analysis of KPIs: Throughput

In this section, the system throughput is evaluated based on varying data sample sizes. As the size of the data samples increases, the number of timeslots required for transmission also grows, which in turn raises the risk of transmission failures. System throughput refers to the amount of data (in bits) received by the destination from the sensors within a given time frame. By analyzing this metric, the relationship between AoI and throughput is investigated. Traditionally, network performance was considered good as long as the system could successfully transmit data, increasing the total amount of data sent per unit of time. However, in recent years, it has become essential to consider the meaning or usefulness of application data, as the scheduling objectives

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now conflict with the conventional performance metric (i.e., system throughput).

Specifically, system throughput decreases mainly due to transmission failures caused by body shadowing for the scheduled sources. As shown in Fig. 13(a), in Test Network A, the performance of the original AoI metric, AoII metric, and combined metric M is generally poor. This is because these metrics base scheduling decisions solely on the AoI of each source as observed from the destination, without integrating dynamic channel conditions. As a result, packet losses lead to a significant decrease in throughput. Although the advanced metric M' incorporates channel conditions into its scheduling decisions and prioritizes sources with higher transmission success probabilities, it does not show a significant performance advantage over the others in environments without clear movement patterns due to the lack of flexible tradeoffs between the age and link quality metrics. On the other hand, as seen in Fig. 13(b), in Test Network B with personalized movement patterns, the advanced metric M' demonstrates the best performance. This is because the coordinator learns the personalized movement patterns and channel statistics of the WBAN user, enabling it to predict future channel conditions and strike a flexible balance between the age metric and link quality.

These results show that scheduling policies dedicated to minimizing AoI are not suitable for system throughput performance in dynamic channel environments, such as WBAN. As previously noted, packet loss not only reduces system throughput but also increases the average AoI. Therefore, it is essential for schedulers in WBAN to consider both the channel conditions and the age of the sources when making scheduling decisions.

H. Discussion

The simulation results reveal several insights regarding the performance of the proposed scheduling scheme across the three KPIs: 1) average AoI; 2) weighted average AoI; and 3) system throughput.

The average AoI analysis shows that while traditional AoI metrics (both the original and AoII) struggle to maintain data freshness in WBANs, especially when body shadowing occurs, the proposed metrics exhibit noticeable improvements. Both the original AoI and AoII metrics fail to adequately handle transmission failures due to their lack of consideration for channel conditions. As expected, in environments with unpredictable body movements (Test Network A), these metrics do not effectively minimize the system's average AoI. The combined metric M slightly improves performance by prioritizing informative data based on variance but still cannot fully prevent AoI increases caused by packet loss. In contrast, the advanced metric M', which integrates both age and link quality, achieves superior results, particularly in Test Network B where personalized movement patterns are present. By predicting future channel conditions, M' successfully mitigates packet loss and reduces the average AoI, demonstrating the importance of incorporating dynamic channel statistics into scheduling decisions.

The weighted average AoI results further emphasize the limitations of traditional AoI-based metrics. In particular, the weighted average AoI captures the freshness of emergency data by considering both the AoI and data deviation. However, simulation results reveal that both the original AoI and AoII metrics are insufficient in maintaining the freshness of informative data, especially in health monitoring systems where data updates are frequent, and the information content between the source and destination may vary significantly. As discussed in the literature review, AoII ultimately penalizes the age in a similar manner to the original AoI metric, resulting in comparable performance between the two. This limitation becomes evident when both metrics struggle to differentiate between regular and highly informative data, which is crucial for timely detection of emergency situations. The combined metric M, however, improves performance by scheduling sources with high data variance, ensuring that more meaningful data are prioritized for transmission. Moreover, the advanced metric M' outperforms all others in scenarios with predictable channel conditions (Test Network B). By balancing age and link quality, M' is able to prioritize both highly informative data and sources with good transmission conditions, thereby minimizing the weighted average AoI. These results suggest that in environments where emergency data detection is critical, scheduling policies that account for both data deviation and channel conditions are necessary for optimal performance.

The throughput analysis reveals an interesting tradeoff between minimizing AoI and maintaining high system throughput. While conventional scheduling policies focused on AoI minimization often overlook channel conditions, leading to higher packet loss and reduced throughput, the advanced metric M' strikes a better balance. In Test Network A, where movements are random, all metrics show similar throughput performance due to the unpredictable nature of channel quality. However, in Test Network B, M' shows significant improvements by learning movement patterns and predicting future channel conditions. These findings demonstrate that while AoI-based metrics are essential for ensuring data freshness, they must be complemented by link-quality-aware scheduling policies to prevent excessive packet loss and resource waste.

VII. Conclusion

To ensure timely and continuous emergency detection in WBAN-based health monitoring systems, this study presented a novel semantic-aware AoI scheduler. The proposed scheduler addressed the limitations of existing semantic-aware AoI schedulers, which often struggled to preserve the freshness of informative data due to their excessive sensitivity to minor variations in information content. To overcome this challenge, the problem of minimizing the age of informative data was reformulated as a weighted average AoI minimization problem. A self-adaptive greedy scheduling algorithm was proposed to achieve this optimization goal. Simulation results indicated that conventional approaches were inadequate for maintaining the freshness of informative data in WBAN-based health monitoring systems. In contrast, the proposed

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scheduler achieved notable improvements across all three KPIs, particularly in scenarios with predictable movement patterns.

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